# Product Requirements Document: Attendora

**Product Type:** Cloud-Based SaaS Web Application (B2B/B2C)
**Industry:** Education Technology (EdTech)
**Document Owner:** Senior Product Manager

## 1. DOCUMENT CONTROL

### Revision History

| Version | Date | Author | Summary of Changes |
|---|---|---|---|
| 1.0 | [Current Date] | Product Manager | Initial Draft - Foundation, Features, Metrics, and MVP Definition |

### Stakeholders & Approvers

| Role | Responsibility | Sign-off Status |
|---|---|---|
| Product Manager | Owns the PRD, prioritization, and product roadmap. | Pending |
| Engineering Lead | Technical feasibility, architecture, and resource allocation. | Pending |
| Design Lead | UI/UX strategy, usability, and visual guidelines. | Pending |
| QA Lead | Test planning, acceptance criteria verification. | Pending |
| Business Stakeholder | Market alignment, budget, and business strategy. | Pending |

**Requirement Priorities Glossary:**
**P0** = Critical for MVP/Launch (Product cannot ship without this).
**P1** = Important/Fast-follow (High value, but launch can proceed conditionally).
**P2** = Desirable/Future phase (Enhancement for future roadmap iterations).

## 2. INTRODUCTION & PRODUCT VISION

### 2.1 Purpose

The purpose of this PRD is to outline the product vision, market context, functional requirements, and technical constraints for **Attendora**. This document will serve as the single source of truth for Engineering, Design, QA, and Business teams throughout the development lifecycle.

### 2.2 Product Vision

To eliminate manual administrative overhead in educational institutions by providing a seamless, secure, and intelligent digital ecosystem for attendance management, enabling educators to focus on teaching and students to take ownership of their academic consistency.

### 2.3 Product Summary

Attendora is a cloud-based SaaS platform that digitizes student attendance tracking, providing real-time data access, automated reporting, and low-attendance alerts via a centralized web portal tailored for students, teachers, and administrators.

### 2.4 Product Scope

**In Scope:** Web-based portals for three user roles (Admin, Teacher, Student), digital attendance marking, cloud storage (Firebase), automated percentage calculations, reporting, and foundational alerting.
**Out of Scope:** Biometric integration (RFID/Fingerprint), native mobile applications (iOS/Android), complex fee management modules, and learning management system (LMS) capabilities.

## 3. PROBLEM STATEMENT & MARKET CONTEXT

### 3.1 The Problem

- **For Teachers:** Marking attendance on paper consumes 10-15% of class time. Manually calculating end-of-month percentages is error-prone and tedious.
- **For Administrators:** Aggregating attendance data across departments is a logistical nightmare, making it difficult to identify at-risk students or generate institutional compliance reports on time.
- **For Students:** Lack of real-time visibility into their attendance status leads to surprise detentions or exam disqualifications at the end of the semester.

### 3.2 Current Problems / Existing Alternatives

Institutions primarily use paper registers, Excel spreadsheets, or legacy on-premise ERP systems that are expensive, difficult to maintain, and lack real-time cloud accessibility. These fragmented systems lead to data silos and a high rate of manual entry errors.

### 3.3 Pain Points

- Data loss due to physical damage or misplacement of paper registers.
- Time-intensive data entry for administrative staff at month-end.
- Delayed intervention for students with chronic absenteeism.

### 3.4 Why Now

With the acceleration of digital transformation in education post-pandemic, institutions are actively seeking lightweight, cloud-native solutions. Regulatory bodies are also demanding stricter compliance and verifiable records for student attendance.

### 3.5 Competitive Landscape

The market contains comprehensive ERPs (which are bloated and expensive) and generic tools (like Google Sheets, which lack specific workflows, role-based access, and automated calculations). Attendora sits in the niche of an affordable, specialized, lightweight SaaS solution.

### 3.6 Product Differentiation

Attendora differentiates through a zero-training user interface, cloud-native architecture enabling instant deployment, and a freemium pricing model that lowers the barrier to entry for smaller institutions.

## 4. GOALS, OBJECTIVES & NON-GOALS

### 4.1 Business Goals

| ID | Goal | Target | Timeframe |
|---|---|---|---|
| BG-01 | Institution Acquisition | Onboard 50 institutions (freemium + paid) | 12 months post-launch |
| BG-02 | Paid Conversion | Convert 15% of freemium institutions to paid tiers | 12 months post-launch |
| BG-03 | User Adoption | 10,000 active daily student/teacher users | Q4 Year 1 |

### 4.2 Product Goals

- Reduce the time taken to mark class attendance to under 60 seconds.
- Provide 100% real-time accuracy on student attendance percentages.
- Ensure a frictionless onboarding experience requiring zero technical intervention.

### 4.3 Non-Goals

- We will not build HR management or payroll modules for teachers.
- We will not support offline-first sync architecture in the initial MVP (active internet is required).

# 5. SUCCESS METRICS & KPIs

## 5.1 North Star Metric

**Daily Attendance Marking Completion Rate:** The percentage of scheduled classes where attendance is digitally logged in the system on the same day. This proves the core value loop (teachers actively preferring the software over paper).

## 5.2 Primary KPIs

| KPI | Definition | Target | Measurement Method |
|---|---|---|---|
| Activation Rate | % of registered teachers who log their first class attendance | > 80% | Product Analytics / DB Logs |
| Time-to-Mark | Average time spent on the attendance screen per session | < 60 seconds | Frontend telemetry |
| DAU/MAU Ratio | Daily Active Users relative to Monthly Active Users | > 60% (weekdays) | Session Tracking |
| System Uptime | Percentage of time the platform is available and operational | 99.9% | Cloud Monitoring (AWS/Render) |

## 5.3 Guardrail Metrics

- **Error Rate:** Must remain below 0.1% for core actions (login, saving attendance).
- **Customer Support Tickets:** Keep under 5 tickets per 100 users monthly to validate UI intuitiveness.

# 6. TARGET USERS & PERSONAS

### Persona A: Priya — The Efficient Educator (Primary User)

- **Role:** Assistant Professor at an Engineering College.
- **Context:** Teaches 3-4 classes a day, each with 60 students.
- **Goals:** Wants to mark attendance quickly and get back to teaching. Needs to download monthly reports for the HOD.
- **Frustrations:** Calling out 60 names takes 10 minutes. Tabulating percentages at month-end ruins her weekend.
- **Device:** Mobile phone (in class) and Laptop (in staff room).

### Persona B: Rahul — The Anxious Student (Secondary User)

- **Role:** 2nd Year Computer Science Student.
- **Context:** Balances classes, projects, and extracurriculars.
- **Goals:** Needs to maintain a 75% attendance criteria to sit for exams.
- **Frustrations:** Doesn't know his exact attendance until the end of the month. Cannot easily prove if he was mistakenly marked absent.
- **Device:** Mobile phone (90% of the time).

### Persona C: Dr. Sharma — The Administrator (Decision Maker)

- **Role:** College Principal / HOD.
- **Context:** Manages institutional compliance, oversees multiple departments.
- **Goals:** Needs a bird's-eye view of departmental attendance and automated defaulter lists.
- **Frustrations:** Relying on faculty to manually compile reports; delayed visibility into chronic absenteeism.

## 7. USER STORIES & JOBS TO BE DONE (JTBD)

### Teachers (Faculty)

- As a teacher, I want to see a list of my scheduled classes for the day, so that I know exactly where to mark attendance.
- As a teacher, I want to mark students present/absent with a single tap, so that I save classroom time.
- As a teacher, I want to generate a PDF report of my class's monthly attendance, so that I can submit it to the administration.

### Students

- As a student, I want to view my overall attendance percentage in real-time, so that I know if I am eligible for exams.
- As a student, I want to receive a notification if my attendance drops below 75%, so that I can take corrective action.

### Administrators

- As an admin, I want to bulk-upload student and teacher rosters via CSV, so that onboarding the institution takes minutes, not days.
- As an admin, I want to view an aggregated dashboard of institution-wide attendance trends, so that I can identify systemic issues.

# 8. SCOPE, ASSUMPTIONS, CONSTRAINTS & DEPENDENCIES

## 8.1 Assumptions

- Users (institutions) will have basic reliable internet access in classrooms.
- Institutions operate on a standard academic calendar model (semesters/years) with defined subjects and enrolled cohorts.
- Students will have access to a smartphone or computer to check their own dashboards.

## 8.2 Constraints

- **Technical Constraints:** Backend built on Python Flask; requires robust WSGI server (Gunicorn) for production. Firebase limitations on concurrent database connections (Free tier limits) dictate data structure optimization.
- **Business Constraints:** Limited initial budget requires prioritizing free/open-source integrations where possible.
- **Compliance:** Must adhere to basic data privacy principles (passwords hashed, PII protected).

## 8.3 Dependencies

| Dependency | Type | Description | Impact if Delayed |
|---|---|---|---|
| Firebase Auth / Firestore | External API | Core database and user authentication provider. | Blocker. Application cannot function without data layer. |
| Cloud Hosting (Render/AWS) | Infrastructure | Environment for deploying Flask application. | Blocker for production launch. |

# 9. FUNCTIONAL REQUIREMENTS

## 9.1 Authentication & Authorization

| ID | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-AUTH-01 | Role-Based Login | P0 | Given a user logs in, the system authenticates credentials against Firebase and routes them to the correct dashboard (Admin, Teacher, Student) based on their assigned role. |
| FR-AUTH-02 | Password Reset | P1 | User can request a password reset link via email, receive it within 2 mins, and successfully update their password securely. |

| ID | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-AUTH-03 | Session Timeout | P1 | User session expires after 24 hours of inactivity, requiring re-authentication. |

### 9.2 Admin Panel & Entity Management

| ID | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-ADM-01 | Create User Accounts | P0 | Admin can manually create a single Student or Teacher profile, assigning unique ID, Name, Email, and Department. |
| FR-ADM-02 | Bulk Upload Users | P1 | Admin can upload a formatted CSV file. System validates schema, creates user accounts, handles errors (e.g., duplicate emails), and outputs a summary report. |
| FR-ADM-03 | Class & Subject Setup | P0 | Admin can create Classes (e.g., "CS 2nd Year") and link specific Subjects and assigned Teachers to them. |

### 9.3 Teacher Attendance Execution

| ID | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-TCH-01 | Select Class/Subject | P0 | Teacher sees a dropdown of assigned subjects. Selecting one displays the roster of enrolled students for the current date. |
| FR-TCH-02 | Mark Attendance | P0 | Teacher can toggle students as Present (default) or Absent. UI clearly distinguishes state. Submit button saves to database with a timestamp. |
| FR-TCH-03 | Edit Attendance | P1 | Teacher can edit an attendance record for up to 48 hours past the class time. After 48h, record is locked and requires Admin override. |

### 9.4 Student Dashboard

| ID | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-STU-01 | View Overall % | P0 | Student dashboard displays aggregate attendance percentage prominently across all subjects. |
| FR-STU-02 | Subject-wise Breakdown | P0 | Student can view a list of subjects with specific attended vs. total classes and the percentage per subject. |
| FR-STU-03 | Historical View | P2 | Student can open a calendar view to see specific days they were marked absent or present. |

### 9.5 Reporting & Notifications

| ID | Requirement | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-REP-01 | Export Defaulter List | P1 | Admin/Teacher can generate a PDF/CSV of students whose attendance is below a specified threshold (e.g., 75%). |
| FR-NOT-01 | Low Attendance Alert | P2 | System automatically triggers an email to the student when their overall percentage drops below institutional requirement. |

## 10. EDGE CASES & EXCEPTION HANDLING

| Feature | Scenario | Expected System Behaviour |
|---|---|---|
| Attendance Submission | Network drops precisely when the teacher clicks "Submit". | UI shows a "Network Error - Retrying" state. Payload is cached locally (session storage) temporarily and pushed upon reconnection, preventing data loss. |
| Duplicate Submission | Teacher double-clicks the submit button rapidly. | Button is disabled immediately upon first click. Backend enforces idempotency, preventing duplicate records for the same class/date combination. |
| CSV Upload | Admin uploads a CSV with missing fields or invalid emails. | Upload transaction aborts. System displays explicit line-by-line error messages (e.g., "Row 5: Invalid email format"). Valid rows are NOT partially committed to ensure data integrity. |
| Late Enrollment | Student joins class 1 month into the semester. | Student is added to roster. System calculates percentage strictly based on total classes held *after* their enrollment date, not the semester start. |

## 11. KEY USER FLOWS

### 11.1 Primary Teacher Flow: Marking Daily Attendance

**Start:** Teacher logs into Web App.
**Action:** Dashboard displays "Today's Schedule". Teacher clicks on specific Subject block.
**Action:** Roster loads. By default, all students are marked "Present".
**Decision:** Teacher identifies absent students and taps their names to toggle status to "Absent".
**Action:** Teacher clicks "Submit Attendance".
**Completion:** Success toast notification appears. Database updates real-time analytics. Teacher is redirected to dashboard.

### 11.2 Administrator Flow: Onboarding a New Class

**Start:** Admin logs in and navigates to "Manage Classes".
**Action:** Clicks "Add New Class", inputs details (Year, Branch, Section).
**Action:** Maps Teacher to the Class and assigns Subjects.
**Decision:** Admin chooses to bulk upload students via CSV.
**Action:** Uploads CSV, maps columns, and clicks "Process".
**Completion:** System validates, creates student accounts, links them to the class, and generates default temporary passwords.

## 12. NON-FUNCTIONAL REQUIREMENTS

| ID | Category | Requirement | Target / Acceptance |
|---|---|---|---|
| NFR-PER-01 | Performance | Roster Load Time | A roster of 100 students must render on the client within 1.5 seconds over a 3G mobile connection. |
| NFR-SEC-01 | Security | Data Protection | All user passwords must be hashed using bcrypt (managed via Firebase Auth). Traffic strictly enforced over HTTPS. |
| NFR-UI-01 | Compatibility | Responsive Design | UI must be fully functional and visually stable on screens as narrow as 320px (mobile portrait) via Bootstrap grid. |
| NFR-SCA-01 | Scalability | Concurrent Users | Backend must handle 500 concurrent teachers submitting attendance at 9:00 AM without request timeouts. |

## 13. INFORMATION ARCHITECTURE & KEY SCREENS

### 13.1 Navigation Hierarchy

- **Admin Panel:** Dashboard (Overview) | Manage Users (Add/Edit) | Manage Classes | Reports | Settings
- **Teacher Panel:** Dashboard (My Classes) | Mark Attendance | My Reports | Profile
- **Student Panel:** Dashboard (Overview Analytics) | Subject Breakdown | Alerts | Profile

### 13.2 Key Screens

| Screen | User | Purpose | Primary Actions |
|---|---|---|---|
| Teacher Dashboard | Teacher | Entry point, schedule view | Select class, View recent submissions |
| Attendance Roster | Teacher | Perform core JTBD | Toggle P/A, Save, Cancel |
| Admin Overview | Admin | Monitor institutional health | View alerts, Export data, Manage entities |

| Screen | User | Purpose | Primary Actions |
|---|---|---|---|
| Student View | Student | Self-service tracking | View %, Check subject details |

## 14. HIGH-LEVEL DATA MODEL

The system utilizes a NoSQL document structure (Firebase Firestore), conceptually mapping to:

- **User Entity:** Contains `uid`, `name`, `email`, `role` (Admin/Teacher/Student), and `institution_id`.
- **Class/Cohort Entity:** Contains `class_id`, `name`, `department`. Contains an array of enrolled `student_uids`.
- **Subject Entity:** Linked to a Class and assigned to a `teacher_uid`.
- **Attendance Record Entity:** A daily log document containing `date`, `subject_id`, `teacher_uid`, and a map/array of `student_uid` statuses (Present/Absent).
  *Relationship:* Teacher -> Creates -> Attendance Record <- References <- Student.

## 15. ROLES & PERMISSIONS

| Role | Resource/Module | Permission |
|---|---|---|
| Administrator | All Users & Classes | Create, Read, Update, Delete |
| Administrator | Global Reports | Read, Export |
| Teacher | Assigned Subject Attendance | Create, Read, Update (within 48h limit) |
| Teacher | Global Users | None (Denied) |
| Student | Own Attendance Data | Read-only |

## 16. INTEGRATIONS & THIRD-PARTY SERVICES

| Integration | Purpose | Criticality | Failure Impact |
|---|---|---|---|
| Firebase Auth | User login, password management, session tokens. | High | Users cannot access the system. |
| Firebase Firestore | Primary database for users, classes, and attendance records. | High | System becomes entirely inoperable. Data cannot be read/written. |
| SendGrid (or similar) | Transactional emails (password resets, low attendance alerts). | Medium | Users cannot reset passwords; alerts delayed. Core attendance logging unaffected. |

## 17. ANALYTICS & INSTRUMENTATION

| Event Name | Trigger | Key Properties | Purpose |
|---|---|---|---|
| `login_success` | User successfully authenticates | `role`, `device_type` | Monitor MAU/DAU and platform engagement. |
| `attendance_submitted` | Teacher clicks save attendance | `class_size`, `time_taken` | Measure North Star metric and UI efficiency. |
| `report_exported` | User downloads PDF/CSV | `role`, `report_type` | Validate value of reporting features. |

## 18. AUDIT & OPERATIONAL LOGGING

To ensure data integrity and resolve disputes, the following actions generate immutable audit logs stored in a dedicated backend collection:

- **Attendance Modification:** Any edit to an attendance record post-submission logs the `teacher_id`, `timestamp`, `student_id` changed, and previous state.
- **User Deletion:** Admin deleting a user logs the action to prevent accidental data loss tracking.
- **Role Changes:** Elevation of a user from Student to Admin (highly restricted) generates a security alert log.

## 19. RELEASE PLAN & ROADMAP

### Phase 1 — MVP (Target: Month 3)

| Phase | Theme | Key Scope | Exit Criteria |
|---|---|---|---|
| Phase 1 | Core Viability | Auth, Manual user creation, Teacher roster view, Basic P/A marking, Student dashboard view. | Stable deployment on Render, capable of supporting 1 test institution for a full month. |
| Phase 2 | Scale & Operations | CSV bulk upload, Automated PDF reports, Advanced admin dashboard, Email alerts. | Onboard 5 institutions. Reduce admin onboarding time to under 1 hour. |
| Phase 3 | Monetization & Enterprise | Multi-department hierarchies, Custom institution branding, API access, Billing module. | Launch paid subscription tiers and acquire first paying B2B customer. |

## 20. MVP DEFINITION

**"The Attendora MVP is complete when a Teacher can log in on their phone, view a pre-configured class roster, mark attendance without errors, and a Student can log in to view their calculated attendance percentage."**

- **Must Have (P0):** Firebase Auth, Teacher marking flow, Student viewing flow, Real-time % calculation, Basic Admin setup UI.
- **Should Have (P1):** CSV bulk upload, Simple CSV export for reports, Edit capabilities within 48h.
- **Later (P2/P3):** Email alerts, PDF generation, Mobile apps, Multi-campus management.

## 21. RISKS & MITIGATIONS

| Risk | Probability | Impact | Mitigation |
|---|---|---|---|
| User Adoption (Teachers prefer paper) | Medium | High | Ensure UI is ultra-fast. The time taken to mark digitally must be demonstrably faster than manual roll-call. |
| Database Scalability Limits (Firebase Free Tier) | High | High | Optimize Firestore read/writes (e.g., store daily attendance as a single document array rather than individual student documents). Upgrade to pay-as-you-go immediately upon exceeding free tier. |
| Data Privacy Compliance | Low | High | Minimize PII collection (only Name, Roll No, Email). Ensure strict Firestore security rules prevent cross-tenant data leakage. |

## 22. OPEN QUESTIONS

| # | Question | Owner | Needed By |
|---|---|---|---|
| 1 | **Pricing Model:** What is the exact pricing tier threshold? (e.g., Free up to 500 students vs. flat module fee). | Business Stakeholder | Phase 3 Start |
| 2 | **Institutional Data Purge:** How long do we retain data after an academic year ends? Do we archive or delete? | Product / Legal | Phase 2 End |
| 3 | **Proxy Attendance Prevention:** Do we need geolocation or network-bound checks in the future to ensure teachers are on campus? | Product / Eng | Future Discovery |

## 23. TRACEABILITY CHECK

**Internal Review Summary:**

- *Business Goal 03 (User Adoption) → KPI (Activation Rate) → JTBD (Teacher marks attendance fast) → FR-TCH-02 → MVP Definition (Must Have).* - **Aligned.**
- *FR-ADM-02 (Bulk Upload)* was marked P1, meaning it is technically acceptable to launch MVP without it, provided admin manual entry is functional. - **Accepted.**
- *Security Rules:* Ensured Firestore security maps directly to Roles & Permissions matrix (Section 15).

## 24. GLOSSARY

- **SaaS:** Software as a Service.
- **B2B / B2C:** Business-to-Business (selling to institutions) / Business-to-Consumer (freemium/student-led adoption).
- **Firebase / Firestore:** Google's cloud-based NoSQL database and authentication suite.
- **Idempotency:** A property of operations in mathematics and computer science whereby an operation can be applied multiple times without changing the result beyond the initial application (crucial for duplicate prevention in submissions).
- **PII:** Personally Identifiable Information (e.g., student names, emails).